Target Tracking by Cameras and Millimeter-Wave Radars: A Confidence Information Fusion Method

Xiaohui Hao, Yuanqing Xia, *Fellow, IEEE*, Hongjiu Yang, and Zhiqiang Zuo, *Senior Member, IEEE*

***Abstract*—This paper addresses a confidence fusion problem of the camera and the millimeter-wave (MMW) radar for target tracking in intelligent driving systems. The local camera and radar estimators are performed by analyzing the measurement characteristics of each sensor. The radar estimates are aligned to the camera sampling time and the Kuhn-Munkers method is used to obtain the matching relationship of local camera and radar estimates for fusion. Next, to utilize the advantage of the camera with low false detection and the radar with low miss detection performance, the mass functions are introduced to model the detection performance of the two sensors. Based on the mass functions and a D-S (Dempster-Shafer) evidence theory, the confidence fusion is performed sequentially to determine whether each target exists. Then a weighted maximum likelihood fusion estimator is designed for matched targets based on priori positing accuracy of the local camera and radar estimates. Finally, the experimental results on road vehicle tracking show that the detection range is expanded and false targets are significantly reduced by the proposed confidence fusion method.**

***Index Terms*—Camera and MMW radar, confidence information fusion, intelligent driving system, target tracking.**

I. Introduction

ENVIRONMENT perception is an important prerequisite to ensure the safety and stability of an intelligent driving system [1]. Sensors commonly used in intelligent driving systems for perception include cameras [2], millimeter-wave radars [3] and lidars [4]. The millimeter-wave radar is abbreviated as radar for conciseness in the following text. The advantages of different sensors are utilized in multi-sensor information fusion to improve perception performance [5]. The camera images usually contain rich information and have wide applications in multiple fields [6], [7], but camera performance is easily affected by lighting and weather [8]–[11]. The radar has high stability, long detection range, and is less affected by the environment, but it cannot recognize the characteristic information of the target. Information fusion of cameras and radars have been studied in some research to improve target tracking performance, as these two sensors are commonly used in practical systems considering positioning performance and cost [12]. One type of camera and radar fusion method is performed at the data level. In [13], the detection results of the radar and the camera are correlated in the image plane and the Gaussian mixture probability hypothesis density algorithm is used to improve the tracking process. A fusion scheme is proposed respectively regarding a radar as the main hardware and a camera as an auxiliary modular in [14]. Target sequences are matched by Mahalanobis distance and fused on the radar module using a joint probability function method. This fusion method can expand the detection range and improve the target detection rate, but it ignores the inconsistent sampling rates of different sensors in practical systems. In addition, the fusion at the data level is usually sensitive to the uncertainty and instability of the sensor data. Subsequently, another type of asynchronous camera and radar fusion method is developed at the track-level. In [15], a fusion estimator of cameras and radars is developed with asynchronous sampling rates. The local camera and local radar estimators are designed based on the camera sensor measurement and the radar sensor measurement, respectively. Then a matrix-weighted fusion algorithm of cameras and radars at synchronized update time is developed to expand the detection range and improve fused estimate accuracy. In [16], an asynchronous fusion estimator of camera and radar is proposed for intelligent driving environment perception based on the Kalman Filter. Specifically, the weight of fusion estimation is constructed based on the matching matrix for local sensor perception estimation. This type of fusion method can also expand the detection range, and fusion at the track level has strong robustness to the errors and noise of each sensor.

Note that the radar obtains target information by electromagnetic wave reflection, and radar detection results often include false targets caused by reflections from objects in roadside areas [17], [18]. In the above two types of fusion methods, the detection range is expanded by supplementing radar results to camera results, but false targets from the radar are also introduced into the fusion results. The false targets are very annoying and may mislead system decisions, increase driving risks, and ultimately affect the safety and stability of the intelligent driving system. Therefore, the purpose of this paper is to simultaneously expand the detection range and reduce false targets by utilizing the complementary detection

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X. Hao is with the School of Artificial Intelligence, Tiangong University, Tianjin 300380, China (e-mail: haoxiaohui@tiangong.edu.cn).

Y. Xia is with the School of Automation, Beijing Institute of Technology, Beijing 100081, and also with Zhongyuan University of Technology, Zhengzhou 450007, China (e-mail: xia_yuanqing@bit.edu.cn).

H. Yang and Z. Zuo are with the School of Electrical and Information Engineering, Tianjin University, Tianjin 300072, China (e-mail: yanghongjiu@tju.edu.cn; zqzuo@tju.edu.cn).

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performance of the camera and the radar.

The meanings of false detection and miss detection are first given to describe sensor characteristics. Target miss detection refers to the fact that the target exists in reality but is not detected by the sensor. The false detection of targets is that the target is mistakenly detected to exist, but it does not exist in reality. The performance of the camera and the radar are complementary by the fact that they obtain target information from image processing and electromagnetic wave reflection, respectively. Note that a camera has a low false detection rate by exclusive vehicle target features in image processing and has a high detection rate for short longitudinal targets with clear target images [19] . As the longitudinal distance increases, the targets become more and more blurred in camera images. The targets with long longitudinal distances are usually missed detection in a camera sensor [20]. A radar has a long longitudinal detection range and low miss detection rate by a principle of electromagnetic wave reflection [21]. The targets in a radar are detected from object's reflections without containing characteristics of vehicle targets. Hence, a large number of false targets are obtained from roadside areas with objects reflected in a radar. After the above analysis, a fusion of the camera and the radar is expected to expand the detection range and reduce false targets utilizing the long detection range of the radar and the low false detection rate of the camera, respectively. Moreover, the different sampling rates and positing accuracy of the camera and the radar should also be considered in information fusion. Therefore, it is interesting to investigate the fusion method to expand the detection range and reduce false targets by considering complementary detection performance, different sampling rates and positing accuracy of the camera and the radar.

In this paper, a confidence fusion scheme is proposed for the camera and the radar in intelligent driving systems. The local camera and local radar estimators, target sequence matching of camera and radar, confidence fusion of target detection results and confidence estimation of matched targets are designed in the confidence fusion estimator. The main contributions of this paper are highlighted as follows:

1) Local camera and radar estimators are designed at their own sampling time based on their measurement characteristics. The radar estimates are aligned to camera sampling time and the Kuhn-Munkers method is used to obtain the matching relationship of local camera and radar estimates for fusion.

2) By fully analyzing the sensor characteristics on miss and false detection performance, the mass functions for the camera and the radar are constructed related to longitudinal and lateral positions, respectively. Then a confidence fusion strategy is developed according to the target detection results and mass functions of the camera and the radar with the D-S (Dempster-Shafer) evidence theory.

3) A weighted maximum likelihood fusion estimator is designed for matched targets by using the respective priori positing accuracy advantages in lateral and longitudinal directions of the camera and the radar.

4) The experimental results in multiple road scenes and poor light conditions show that the detection range is expanded and false targets are significantly reduced by the proposed confidence fusion method compared with the recent works.

Notations: Most notations used in this paper are standard. In this paper, $\mathcal{N}(x;m,P)$ denotes the normal distribution, where m and P are the mean and covariance of the variable x, respectively, $\cup$ is a symbol of union operation, $\max\{a,b\}$ represents the larger value between a and b, $\mathrm{diag}\{\cdot\}$ denotes a diagonal matrix, $\mathbb{E}\{\cdot\}$ represents the expected operation, I_s represents the unit matrix with $s\times s$ dimension.

II. Problem Formulation

A. Experimental Platform

An intelligent driving vehicle in this paper is shown in Fig. 1, which includes a camera and a radar in the perceptive modular. The camera model is BFLY-PGE-23S6C-C and the sampling rate is 30 Hz. Bounding boxes and identifications (IDs) of vehicle targets are acquired by a DeepSORT-YOLO algorithm from camera images and target positions are acquired by a perspective transformation [22] . The radar model is ARS408-21 and the sampling rate is 10 Hz. The target positions and velocities are obtained by electromagnetic wave reflection in the radar.



Fig. 1. The perception modular of the intelligent driving vehicle.

Detection ranges of the sensors are shown in Fig. 2, in which the camera and radar detection ranges are presented by the blue and orange fans, respectively. Note that the radar detection range with a long longitudinal position is complementary to camera detection range. However, radar detection results may contain roadside obstacles for the reason that characteristics of vehicle targets are not included in the electromagnetic wave reflection process. For example, the circle target in Fig. 2 may be a roadside obstacle by the fact that the circle target is not detected by the camera in its effective detection area. The roadside obstacles in the detection results produce false alarms in the sensing system, which will affect the decision-making of intelligent vehicles. By integrating complementarity of the camera and the radar, a sensing system is expected to expand detection range, reduce false target numbers, and improve tracking accuracy in intelligent vehicles.

Remark 1: The radar ARS408-21 is a 77 GHz sensor with a digital beam-forming scanning antenna which offers two independent scans for far and near range. During one measurement cycle, one far range scan of ±9° with a distance range of 0.25 – 250 m and one near range scan of ±60° with a distance range of 0.25 – 70 m is done. The detection range is discontinuous of the radar ARS408-21 by the fact that the far range and

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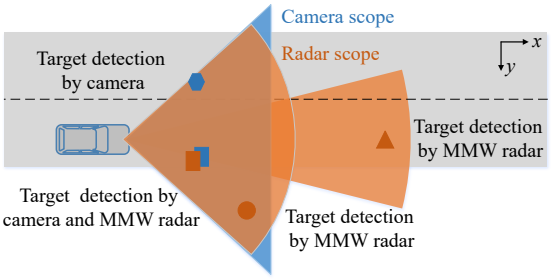


Fig. 2. Detection ranges and targets of the camera and MMW radar.

near range have different detection distances. The detection range of the radar in Fig. 2 is a schematic diagram based on the detection range of ARS408-21, in which two sectors represent the near range and the far range, respectively.

B. System Model of the Camera Sensor

In the camera sensor, the target IDs and positions are obtained from the camera images. Then target motion model and measurement model are described for each target independently based on the target IDs. In [23], a target motion model for tracking targets is presented as

$$\xi_{l,c}(k_c+1)=F_{l,c}(k_c)\xi_{l,c}(k_c)+w_{l,c}(k_c) \tag{1}$$

$$F_{l,c}(k_c)=\mathcal{F}(T_c) \tag{2}$$

$$\mathcal{F}(T_c)=\begin{bmatrix}1 & T_c & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & T_c\\ 0 & 0 & 0 & 1\end{bmatrix},\ l=1,2,\ldots,N_c(k_c) \tag{3}$$

where $\xi_{l,c}(k_c)=[x_{l,c}(k_c)\ \vartheta_{x,l}(k_c)\ y_{l,c}(k_c)\ \vartheta_{y,l}(k_c)]^T$ is the target state in which $x_{l,c}(k_c)$ and $\vartheta_{x,l}(k_c)$ are target position and velocity in longitudinal direction, respectively, $y_{l,c}(k_c)$ and $\vartheta_{y,l}(k_c)$ are target position and velocity in lateral direction, respectively, k_c and T_c are the state beat and sampling time of the camera, respectively, $N_c(k_c)$ is the target number at beat k_c, the process noise $w_{l,c}(k_c)$ is white and zero-mean with covariance $Q_{l,c}$. The camera measurement model of target l is obtained as

$$z_{l,c}(k_c)=H_{l,c}(k_c)\xi_{l,c}(k_c)+v_{l,c}(k_c) \tag{4}$$

$$H_{l,c}(k_c)=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix} \tag{5}$$

where $z_{l,c}(k_c)$ is the camera measurement of target l at time beat k_c, $v_{l,c}(k_c)$ is zero-mean and white with known covariance $R_{l,c}$. Target measurements are nonexist in the case that targets are undetected by the camera. A binary function $\gamma_c(k_c)$ is introduced to indicate whether target measurement is displayed or not of the camera, in which $\gamma_c(k_c)=1$ or $\gamma_c(k_c)=0$ represents that the target is detected or undetected, respectively.

C. System Model of the Radar Sensor

A radar sensor acquires target information by electromagnetic wave reflection. Target measurements and clutters from other obstacles are obtained forming disorder radar measurements. Note that an estimator based on a random finite set (RFS) is an effective way for multiple target tracking with clutters [24]. Target states and measurements in the radar sensor are modeled by the state RFS $\Xi_r(k_r)$ and the measurement RFS $Z_r(k_r)$ as follows:

$$\Xi(k_r)=\{\xi_1(k_r),\ldots,\xi_{N_r(k_r)}(k_r)\} \tag{6}$$

$$Z_r(k_r)=\{z_{r,1}(k_r),\ldots,z_{r,M_r(k_r)}(k_r)\} \tag{7}$$

where k_r is the state update beat of the radar, $N_r(k_r)$ and $M_r(k_r)$ are targets and measurements number at state update beat k_r, respectively. The dynamic model for state $\Xi(k_r)$ is denoted as

$$\Xi(k_r)=\left[\cup_{\xi_r(k_r-1)\in\Xi_r(k_r-1)}S_{k_r|k_r-1}(\xi_r(k_r-1))\right]\cup B(k_r)$$

where $S_{k_r|k_r-1}(\xi_r(k_r-1))=\emptyset$ if the target dies or $S_{k_r|k_r-1}(\xi_r(k_r-1))=\{\xi_r(k_r)\}$ if the target survives, $B(k_r)$ is the RFS of birth targets. The motion model of the target is

$$\xi_r(k_r+1)=F_r(k_r)\xi_r(k_r)+w_r(k_r) \tag{8}$$

where $F_r(k_r)=\mathcal{F}(T_r)$, T_r is the sampling period of the radar, the process noise $w_r(k_r)$ is white and zero-mean with a known covariance Q_r. The multiple target measurements $Z_r(k_r)$ from the state $\Xi_r(k_r)$ is described as

$$Z_r(k_r)=\left[\cup_{\xi_r(k_r)\in\Xi_r(k_r)}\Theta(\xi_r(k_r))\right]\cup\Upsilon(k_r)$$

where $\Theta(\xi_r(k_r))=\emptyset$ if the target is not detected or $\Theta(\xi_r(k_r))=\{z_r(k_r)\}$ if the target is detected. $\Upsilon(k_r)$ is the clutters Poisson RFS. The target measurement $z_r(k_r)$ is determined by

$$z_r(k_r)=H_r(k_r)\xi(k_r)+v_r(k_r) \tag{9}$$

$$H_r(k_r)=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix} \tag{10}$$

where $v_r(k_r)$ is white and zero-mean with covariance R_r.

III. Confidence Fusion of the Camera and the Radar

A confidence fusion estimator is designed in this section based on detection and positioning performance of the camera and the radar. Fusion processes are shown in Fig. 3, which include the local estimation, the target matching and the confidence fusion. The local estimators of the camera and radar sensors are presented firstly by system models (1)−(5) and (6)−(9), respectively. Then time stamp alignment and target sequence matching of the camera and the radar are performed to obtain target matching results. Finally, a confidence fusion strategy is developed for target detection results of the camera and the radar and confidence estimation of matched targets is designed. The implementation details are described in the following subsections.

A. Local Estimator of the Camera Sensor

A local estimator of camera sensor is developed by using the target motion models (1)−(5) and Kalman filter structure. Note that the estimators with a Kalman filter structure can effectively suppress the measurement noises [25]–[28]. The

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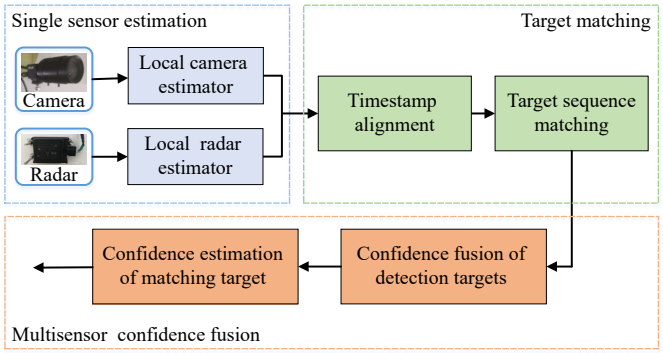


Fig. 3. Confidence information fusion of the camera and radar.

predicted estimates $\hat{\xi}_{l,c}(k_c|k_c-1)$ and covariance $P_{l,c}(k_c|k_c-1)$ for local camera sensor at beat k_c is given as

$$\hat{\xi}_{l,c}(k_c|k_c-1) = F_{l,c}(k_c-1)\hat{\xi}_{l,c}(k_c-1) \tag{11}$$

$$P_{l,c}(k_c|k_c-1) = F_{l,c}(k_c-1)P_{l,c}(k_c-1)F_{l,c}^T(k_c-1) + Q_{l,c} \tag{12}$$

Note that target IDs are obtained from camera images in target detection process. Target measurements with IDs are used in state update estimates of each target. The updated estimates $\hat{\xi}_{l,c}(k_c)$ and covariance $P_{l,c}(k_c)$ are obtained as follows:

$$\zeta_{l,c}(k_c) = z_{l,c}(k_c) - H_{l,c}(k_c)\hat{\xi}_{l,c}(k_c|k_c-1) \tag{13}$$

$$S_{l,c}(k_c) = H_{l,c}(k_c)P_{l,c}(k_c|k_c-1)H_{l,c}^T(k_c) + R_{l,c} \tag{14}$$

$$K_{l,c}(k_c) = P_{l,c}(k_c|k_c-1)H_{l,c}^T(k_c)S_{l,c}^{-1}(k_c) \tag{15}$$

$$\hat{\xi}_{l,c}(k_c) = \hat{\xi}_{l,c}(k_c|k_c-1) + \gamma_{l,c}(k_c)K_{l,c}(k_c)\zeta_{l,c}(k_c) \tag{16}$$

$$P_{l,c}(k_c) = (I - \gamma_{l,c}(k_c)K_{l,c}(k_c)H_{l,c}(k_c))P_{l,c}(k_c|k_c-1) \tag{17}$$

where $\gamma_{l,c}(k_c)=1$ or $\gamma_{l,c}(k_c)=0$ represents that there exist or does not exist measurements at beat k_c, respectively. Note that the updated estimates and covariance in (16) and (17) are assigned with predicted ones in the case that targets are undetected by the camera with $\gamma_c(k_c)=0$.

B. Local Estimator of the Radar Sensor

Based on the RFS system models (6)−(9), the local estimator of the radar sensor is obtained by a Gaussian mixture-probability hypothesis density (GM-PHD) filter which is a widely used filter in RFS framework with low computational complexity [29]. The steps of the GM-PHD filter are listed as follows:

1) Predicted Process: Suppose that the posterior intensity at beat k_r-1 is a Gaussian mixture with the form

$$D(\xi_r, k_r-1) = \sum_{i=1}^{J(k_r-1)} \omega_i(k_r-1)\mathcal{N}(\xi_r;\rho_i(k_r-1), P_i(k_r-1))$$

where $J(k_r-1)$ is the number of Gaussian components at beat k_r-1, $\omega_i(k_r-1)$, $\rho_i(k_r-1)$ and $P_i(k_r-1)$ are weight, mean and covariance of the Gaussian component i, respectively.

The predicted intensity for beat k_r is given by

$$D(\xi_r, k_r|k_r-1) = D_S(\xi_r, k_r|k_r-1) + D_B(\xi_r, k_r) \tag{18}$$

where

$$D_S(\xi_r, k_r|k_r-1) = p_S \sum_{i=1}^{J(k_r-1)} \omega_i(k_r-1) \times \mathcal{N}(\xi_r;\rho_{i,S}(k_r|k_r-1), P_{i,S}(k_r|k_r-1))$$

$$\rho_{i,S}(k_r|k_r-1) = F_r(k_r-1)\rho_i(k_r-1)$$

$$P_{i,S}(k_r|k_r-1) = Q_r + F_r(k_r-1)P_i(k_r-1)F_r^T(k_r-1)$$

$$D_B(\xi_r, k_r) = \sum_{i=1}^{J_B(k_r)} \omega_{i,B}(k_r)\mathcal{N}(\xi_r;\rho_{i,B}(k_r), P_{i,B}(k_r))$$

where p_S is the survives probability. The predicted intensity (18) consists of existing targets $D_S(\xi_r, k_r|k_r-1)$ and birth targets $D_B(\xi_r, k_r)$ two parts. The prediction of existing targets $D_S(\xi_r, k_r|k_r-1)$ is calculated based on the previous target track information. The intensities of the birth targets $D_B(\xi_r, k_r)$ are expressed as Gaussian mixture form, in which $J_B(k_r)$ is the number of Gaussian components of the birth targets, $\rho_{i,B}(k_r)$ is the peaks of the birth intensity, the covariance matrix $P_{i,B}(k_r)$ determines the spread of the birth intensity in the vicinity of the peak $\rho_{i,B}(k_r)$ and the weight $\omega_{i,B}(k_r)$ gives the expected number of new targets originating from $\rho_{i,B}(k_r)$.

2) Updated Process: Suppose that the predicted intensity at beat k_r is a Gaussian mixture with the following form:

$$D(\xi_r, k_r|k_r-1) = \sum_{i=1}^{J(k_r|k_r-1)} \omega_i(k_r|k_r-1) \times \mathcal{N}(\xi_r;\rho_i(k_r|k_r-1), P_i(k_r|k_r-1)).$$

The posterior intensity at beat k_r is given by

$$D(\xi_r, k_r) = \sum_{z\in Z_r(k_r)} z_D(k_r) + (1-p_D)D(\xi_r, k_r|k_r-1) \tag{19}$$

where

$$z_D(k_r) = \sum_{i=1}^{J(k_r|k_r-1)} \omega_{i,z}(k_r)\mathcal{N}(\xi_r;\rho_{i,z}(k_r), P_{i,z}(k_r))$$

$$\omega_{i,z}(k_r) = \frac{p_D\omega_i(k_r|k_r-1)q_{i,z}(k_r)}{C_z(k_r) + \sum_{j=1}^{J(k_r|k_r-1)} \omega_j(k_r|k_r-1)q_{j,z}(k_r)}$$

$$q_{i,z}(k_r) = \mathcal{N}(z_r; H_r(k_r)\rho_i(k_r|k_r-1), H_r(k_r)P_i(k_r|k_r-1)H_r^T(k_r) + R_r)$$

$$\rho_{i,z}(k_r) = \rho_i(k_r|k_r-1) + K_{i,r}(k_r) \times (z_r - H_r(k_r)\rho_i(k_r|k_r-1))$$

$$K_{i,r}(k_r) = P_i(k_r|k_r-1))H_r^T(k_r)(H_r(k_r)P_i(k_r|k_r-1)) \times H_r^T(k_r) + R_r)^{-1}$$

$$P_{i,z}(k_r) = (I - K_{i,r}(k_r)H_r(k_r))P_i(k_r|k_r-1)$$

where p_D is the detection probability, $C_z(k_r)$ is the clutter

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intensity at time k_r. The updated posterior intensity $D(\xi_r, k_r)$ in (19) consists of detection term $z_D(k_r)$ and a misdetection term $(1-p_D)D(\xi_r, k_r|k_r-1)$. The recursions for the mean $\rho_{i,z}(k_r)$ and covariance $P_{i,z}(k_r)$ of detection term $z_D(k_r)$ are calculated by Kalman update formulas. Based on the posterior intensity (19), target number $N_r(k_r)$, target estimates $\hat{\xi}_{j,r}(k_r)$ $(j=1,2,\ldots,N_r(k_r))$ and corresponding estimation covariance $P_{j,r}(k_r)$ are able to obtain by pruning, merging as well as state extraction steps. For the detailed process, please refer to (44) and (45), and Tables II and III in [29].

C. Target Matching

The local target tracking results are obtained by local estimators of the camera and radar at each beat. The time stamp of the local radar tracking results is aligned with the local camera results for fusion. The different sampling times of the camera and radar are presented in Fig. 4. The time interval of k_r radar beat to the $(3k_r-2)$, $(3k_r-1)$ and $(3k_r)$ camera beats are T_1, T_2 and T_3, respectively. The time interval T_i $(i=1, 2, 3)$ are constants in the whole process by the fact that the sampling rates of the radar and camera are fixed. The radar estimation results are predicted by T_i $(i=1,2,3)$ time interval to align the sampling times of the camera estimator. The radar estimation results of target j at each camera update beat k_c are obtained as

$$\hat{\xi}_{j,r}(k_c)=\begin{cases}\mathcal{F}(T_1)\hat{\xi}_{j,r}(k_r), & \text{if } k_c=3k_r-2\\ \mathcal{F}(T_2)\hat{\xi}_{j,r}(k_r), & \text{if } k_c=3k_r-1\\ \mathcal{F}(T_3)\hat{\xi}_{j,r}(k_r), & \text{if } k_c=3k_r.\end{cases} \tag{20}$$

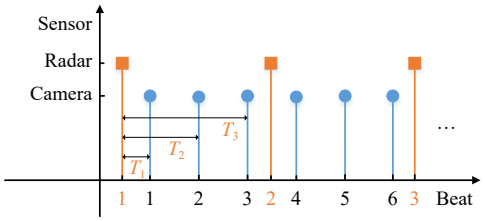


Fig. 4. The different sampling rates of camera and radar.

Next the local camera and radar estimates are matched by a Kuhn-Munkers algorithm at each beat k_c. The camera target number and radar target number are $N_c(k_c)$ and $N_r(k_c)$ at beat k_c, respectively. The matching degree of camera estimates $\hat{\xi}_{\alpha,c}(k_c)$ and radar estimates $\hat{\xi}_{\beta,r}(k_c)$ are presented by a weight matrix $W^f(k_c)$ with $N_f(k_c)\times N_f(k_c)$ dimensions in which $N_f(k_c)=\max\{N_c(k_c), N_r(k_c)\}$. Denoting $w^f_{\alpha\beta}(k_c)$ is the (α,β)th element of the weight matrix $W^f(k_c)$. The (α,β)th element $w^f_{\alpha\beta}(k_c)$ is constructed as

$$w^f_{\alpha\beta}(k_c)=\begin{cases}0, & \text{if } \alpha>N_c(k_c) \text{ or } \beta>N_r(k_c)\\ 0, & \text{if } D^f_{\alpha\beta}(k_c)>d^f\\ 30-D^f_{\alpha\beta}(k_c), & \text{otherwise}\end{cases} \tag{21}$$

where $D^f_{\alpha\beta}(k_c)=\|\hat{\xi}_{\alpha,c}(k_c)-\hat{\xi}_{\beta,r}(k_c)\|_2$ denotes the distance of camera estimates $\hat{\xi}_{\alpha,c}(k_c)$ and radar estimates $\hat{\xi}_{\beta,r}(k_c)$, $0<d^f<30$ is a threshold to reject matching. In (21), if camera estimates $\hat{\xi}_{\alpha,c}(k_c)$ $(\alpha>N_c(k_c))$ are not exist or radar estimates $\hat{\xi}_{\beta,r}(k_c)$ $(\beta>N_r(k_c))$ are not exist, we have $w^f_{\alpha\beta}(k_c)=0$. Otherwise, the matching degree is related to the distance $D^f_{\alpha\beta}(k_c)$ of camera estimates and radar estimates. Note that the matching degree decreases as the distance $D^f_{\alpha\beta}(k_c)$ increases. In the case of $D^f_{\alpha\beta}(k_c)>d^f$, it is considered that camera estimates $\hat{\xi}_{\alpha,c}(k_c)$ and radar estimates $\hat{\xi}_{\beta,r}(k_c)$ have no matching relationship and we have $w^f_{\alpha\beta}(k_c)=0$. The matching degree $w^f_{\alpha\beta}(k_c)=30-D^f_{\alpha\beta}(k_c)$ in the third line of (21) ensures that $w^f_{\alpha\beta}(k_c)$ is a positive value and decreases as $D^f_{\alpha\beta}(k_c)$ increases. Based on the weight matrix $W^f(k_c)$, matching relations of local camera and radar estimates are acquired by a Kuhn-Munkers algorithm.

D. Confidence Fusion of the Target Detection Results

The fusion of target detection results is the process of obtaining the fused target list based on camera and radar target detection results. For example, the radar detects a target at a certain location but the camera does not detect it. A decision needs to be made by information fusion about whether there exists a target at this location based on the camera and radar detection results. The camera and radar have their detection confidence in different areas by the fact that detection principles are inconsistent. Fusion of target detection results should fully consider detection confidences of the camera and radar to take advantage of their respective strengths. In this paper, confidence fusion is performed by a D-S evidence theory method by quantifying the detection mass degree of camera and radar sensors [30]. Herein, the recognition framework is denoted as $\Theta=\{A,\bar{A}\}$ in which A and $\bar{A}$ represent the target existence event and target nonexistence event, respectively. The power set 2^Θ of recognition framework Θ is obtained as

$$2^\Theta=\{A,\bar{A},\{A,\bar{A}\},\phi\} \tag{22}$$

where ϕ is an empty set. The power set 2^Θ contains target existence event A, target nonexistence event $\bar{A}$, uncertain event $\{A,\bar{A}\}$ or an empty set ϕ. The meaning of $\{A,\bar{A}\}$ represents an uncertain event, that is, whether the target exists or does not exist is uncertain. The basic probability assignment function m satisfies two conditions as follows:

$$\{m(\phi)=0, \sum_{B\subseteq 2^\Theta} m(B)=1\} \tag{23}$$

where $m(B)$ is called mass function of B, which indicates confidence of subset B. Next mass functions are designed for the camera and radar sensors.

The camera detects targets from the image, but the target becomes blurred in the image as the target longitudinal distance increases. It is shown that camera detection performance is related to the target longitudinal position. The camera mass function $m_1(\cdot)$ is given in Fig. 5 with the increase of longitudinal position. Denote " Detected target case" or

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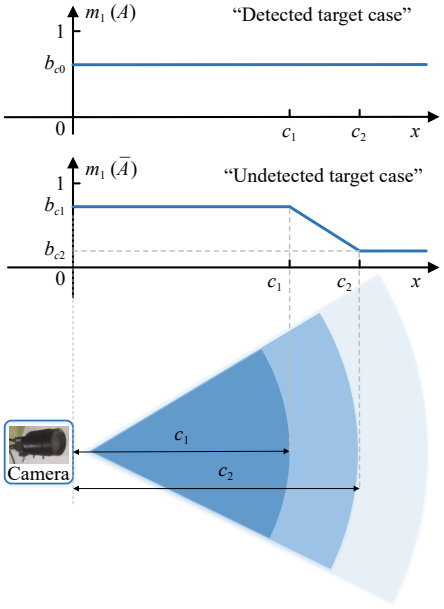


Fig. 5. Confidence value of the camera in the detection area.

"Undetected target case" is the situation that a target is detected or undetected by the camera, respectively. Next the mass values of the "Detected target case" and "Undetected target case" are designed for the camera sensor. The mass functions $m_1(A)$ and $m_1(\bar{A})$ of the camera represent the existence confidence and nonexistence confidence of the target, respectively. The parameter c_1 represents the maximum longitudinal distance in which the camera has stable detection performance. The parameter c_2 indicates the maximum value of the longitudinal detection range and the camera sensor will most likely not be able to detect the vehicle if the longitudinal distance of the target vehicle in the range $\hat{x}_c(k_c) > c_2$. Note that the camera detects vehicle targets by exclusive features of the vehicle, and vehicle targets detected by the camera indicate the target existence event A with high confidence. If a vehicle target is detected in a camera image, it is highly likely to indicate the true existence of the vehicle target, as the camera detection process is based on the exclusive features of the vehicle. The mass value of target existence event A in the "Detected target case" is

$$m_1(A) = b_{c0} \tag{24}$$

and b_{c0} is a constant close to 1, which indicates that the target existence event has a high confidence degree. By the fact that targets become more and more blurred in images as the target longitudinal distance increases. The detection performance of the camera decreases as the target longitudinal distance increases, which means that the camera is prone to miss targets with large longitudinal positions. Hence, in the "Undetected target case", the confidence degree of the target nonexistence event $\bar{A}$ decreases as the longitudinal distance increases. The mass value of the "Undetected target case" for target nonexistence event $\bar{A}$ is $m_1(\bar{A}) = b_c$ which is given by

$$m_1(\bar{A}) = b_c = \begin{cases} b_{c1}, & \hat{x}_c(k_c) \le c_1 \\ b_c(\hat{x}_c(k_c)), & c_1 < \hat{x}_c(k_c) \le c_2 \\ b_{c2}, & \hat{x}_c(k_c) > c_2 \end{cases} \tag{25}$$

where $b_c(\hat{x}_c(k_c)) = \frac{(b_{c1}-b_{c2})\hat{x}_c(k_c)+c_1 b_{c2}-c_2 b_{c1}}{c_1-c_2}$. Note that $\hat{x}_c(k_c) \le c_1$ is the stable detection range of the camera and the undetected targets in the camera indicates the target nonexistence event with high confidence. Hence, the confidence degree b_{c1} is also a large value close to 1. For area $\hat{x}_c(k_c) > c_2$, the detection performance of the camera decreases and the probability of the target being missed is high. The undetected target in the camera indicates the target nonexistence event with low confidence in area $\hat{x}_c(k_c) > c_2$, and b_{c2} is a value close to 0. The mass values for the camera sensor are summarized in Table I containing " Detected target case" and " Undetected target case".

TABLE I
THE MASS VALUE FOR CAMERA SENSOR

Mass value	$m_1(A)$	$m_1(\bar{A})$	$m_1(\{A,\bar{A}\})$	$m_1(\phi)$
Detected target case	b_{c0}	0	$1-b_{c0}$	0
Undetected target case	0	b_c	$1-b_c$	0

The radar detects targets by electromagnetic wave reflection, and obstacles in roadside areas are included in the detection results. It is shown that radar detection performance is related to the lateral position. Confidence of radar target detection results is presented in lateral dimension which is shown in Fig. 6. Next the mass values of the "Detected target case" and "Undetected target case" are designed for the radar sensor. The mass functions $m_2(A)$ and $m_2(\bar{A})$ of the radar represent the existence confidence and nonexistence confidence of the target, respectively. In the "Detected target case", due to the low miss detection rate of the radar, the detected target of the radar in road area $r_2 < \hat{y}_r(k_c) < r_3$ indicates the target existence event A with high confidence and b_{r1} is a value close to 1. However, false targets are contained in radar detection results in the roadside areas $\hat{y}_r(k_c) < r_4$ and $\hat{y}_r(k_c) > r_1$. The confidence degree decreases as the lateral position extends to the roadside and b_{r2} is a value close to 0. The mass function $m_2(A)$ in the "Detected target case" of radar is presented as

$$m_2(A) = b_r = \begin{cases} b_{r2}, & \hat{y}_r(k_c) \le r_1 \\ b_{r12}(\hat{y}_r(k_c)), & r_1 < \hat{y}_r(k_c) \le r_2 \\ b_{r1}, & r_2 < \hat{y}_r(k_c) \le r_3 \\ b_{r34}(\hat{y}_r(k_c)), & r_3 < \hat{y}_r(k_c) \le r_4 \\ b_{r2}, & \hat{y}_r(k_c) > r_4 \end{cases} \tag{26}$$

where $b_{r12}(\hat{y}_c(k_c)) = \frac{(b_{r1}-b_{r2})\hat{y}_c(k_c)+r_2 b_{r2}-r_1 b_{r1}}{r_2-r_1}$, $b_{r34}(\hat{y}_c(k_c)) = \frac{(b_{r1}-b_{r2})\hat{y}_c(k_c)+r_3 b_{r2}-r_4 b_{r1}}{r_3-r_4}$. The radar mass function $m_2(A)$ reflects the fact that the more the lateral distance extends to roadsides, the more likely the detected target is a roadside obstacle. In the "Undetected target case", the undetected target of the radar indicates the target nonexistence event $\bar{A}$ with high confidence due to the low miss detection rate of the

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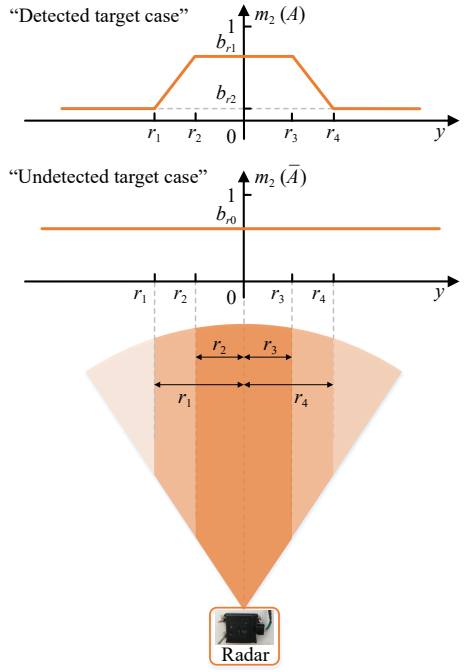


Fig. 6. Confidence value of the radar in the detection area.

radar. If there is no vehicle target in a certain position, it is highly likely that there is no target at that location in the radar results. The mass value of target nonexistence event $\bar{A}$ in the "Undetected target case" is

$$m_2(\bar{A}) = b_{r0} \tag{27}$$

and b_{r0} is a constant close to 1, which indicates that the target nonexistence event has a high confidence degree. The mass values in the "Detected target case" and "Undetected target case" for the radar sensor are summarized in Table II.

TABLE II
THE MASS VALUE FOR RADAR SENSOR

Mass value	$m_2(A)$	$m_2(\bar{A})$	$m_2(\{A,\bar{A}\})$	$m_2(\phi)$
Detected target case	b_r	0	$1-b_r$	0
Undetected target case	0	b_{r0}	$1-b_{r0}$	0

Based on matching results of the camera and radar, the camera mass function $m_1(\cdot)$ is determined from "Detected target" or "Undetected target" cases in Table I, and the radar mass function $m_2(\cdot)$ is determined from "Detected target" or "Undetected target" cases in Table II. The orthogonal sum to combine two pieces of evidence is calculated by Dempster's combination rule as follows:

$$m(B) = \frac{1}{1-\kappa} \sum_{B_1 \cap B_2 = B} m_1(B_1) m_2(B_2) \tag{28}$$

$$\kappa = \sum_{B_1 \cap B_2 = \phi} m_1(B_1) m_2(B_2),\ B \subseteq 2^{\Theta}, B \neq \phi \tag{29}$$

where $m(B) = (m_1 \oplus m_2)(B)$. Then judgment criteria are given by the combined mass value $m(B)$. If there exist $B_1, B_2 \subseteq 2^{\Theta}$ satisfying

$$m(B_1) = \max\{m(B_i), B_i \subseteq 2^{\Theta}\} \tag{30}$$

$$m(B_2) = \max\{m(B_i), B_i \subseteq 2^{\Theta}, B_i \neq B_1\} \tag{31}$$

then B_1 is the identification result if following inequalities:

$$m(B_1) - m(B_2) > \varepsilon_1 \tag{32}$$

$$m(\{B_1, B_2\}) < \varepsilon_2 \tag{33}$$

$$m(B_1) > m(\{B_1, B_2\}) \tag{34}$$

hold, where the event with the largest mass value is recorded as B_1, and the event with the second largest mass value is recorded as B_2, parameters ε_1 and ε_2 are thresholds defined by a user.

Remark 2: The events B_1 and B_2 are elements in power set 2^{Θ} given by (22). Specifically, B_1 and B_2 denote target existence event A, target nonexistence event $\bar{A}$, uncertain event $\{A,\bar{A}\}$ or an empty set ϕ. In the Dempsters combination rules (28) and (29), $m_1(B_1)$ and $m_2(B_2)$ are determined by camera mass values in Table I and radar mass values in Table II, respectively. In the judgment criterions (32)–(34), B_1 and B_2 represent events with maximum mass value $m(B_1)$ and secondary mass value $m(B_2)$, respectively. The event B_1 is the fusion result of camera and radar target detection results if the judgment criterions inequalities (32)–(34) hold.

To clarify, a detailed explanation of the process of detection results fusion is shown in Fig. 7. In Fig. 7, three types of targets are obtained by target matching. The fusion of detection results is performed based on these three types of targets and the mass values in Tables I and II, which are divided into the following three cases.

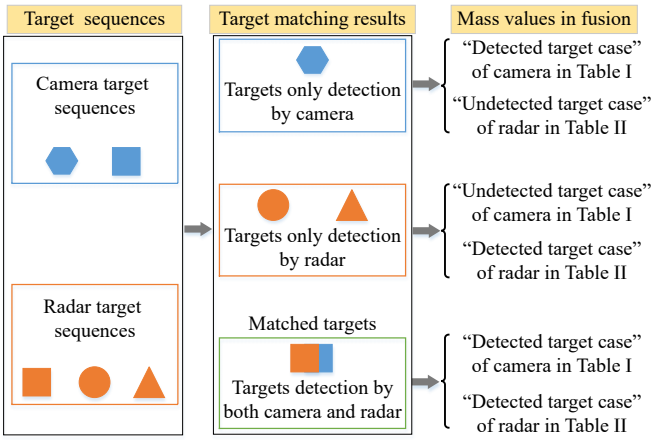


Fig. 7. Confidence fusion of target detection results.

Case 1: Targets only detection by the camera. This means that the target is detected by the camera but not detected by the radar. The mass values of the "Detected target case" in the camera and the mass values of the "Undetected target case" in the radar are used for fusion.

Case 2: Targets only detection by the radar. This means that the target is detected by the radar but not detected by the camera. The mass values of the "Detected target case" in the radar and the mass values of the "Undetected target case" in the

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camera are used for fusion.

Case 3: Targets detection by both camera and radar. This means that the target is detected by the camera and radar simultaneously. The mass values of the "Detected target case" in the camera and the mass values of the " Detected target case" in the radar are used for fusion.

Remark 3: For the targets only detection by the radar (or the camera), the mass values of the "Detected target case" in the radar (or the camera) and the mass values of the "Undetected target case" in the camera (or the radar) are used for fusion. There is no target location information in the camera (or the radar) by the fact that the target is not detected by the camera (or the radar). Hence, the target estimation information from the radar (or the camera) is used in the mass value calculation of the "Undetected target case" for the camera (or the radar) in Table I (or Table II).

E. Confidence Estimation of Matching Targets

The states of matched targets are obtained by fusing local estimates of the camera and radar to improve estimation accuracy. Local camera and radar estimates have advantages in lateral and longitudinal position, respectively. A weighted maximum likelihood estimation is used to leverage strengths of local camera and radar estimates. Fused estimate $\hat{\xi}_f(k_c)$ is obtained as

$$\hat{\xi}_f(k_c) = \arg\max_{\xi} \frac{1}{\gamma} e^{-\frac{1}{2}G(k_c)} \tag{35}$$

where

$$\gamma = 4\pi^2 |P_c(k_c)|^{\frac{1}{2}} |P_r(k_c)|^{\frac{1}{2}}$$

$$G(k_c) = \|C_c(\xi - \hat{\xi}_c(k_c))\|^2_{P_c(k_c)} + \|C_r(\xi - \hat{\xi}_r(k_c))\|^2_{P_r(k_c)}$$

$$C_c = \begin{bmatrix} w_{c,x} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1-w_{r,y} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_r = \begin{bmatrix} 1-w_{c,x} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & w_{r,y} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\hat{\xi}_c(k_c)$ and $P_c(k_c)$ are target estimates and its covariance obtained by the camera sensor, respectively, $\hat{\xi}_r(k_c)$ and $P_r(k_c)$ are target estimates and its covariance obtained by the radar sensor, respectively, $|P_c(k_c)|$ and $|P_r(k_c)|$ denote the determinant of covariance matrices $P_c(k_c)$ and $P_r(k_c)$, respectively. Parameters $w_{c,x}$ and $w_{r,y}$ are weights of camera in longitudinal direction and radar in lateral direction, respectively, the symbol $\|\cdot\|^2_{P_c(k_c)}$ denotes the weighted norm. By the fact that areas with low confidence should be assigned low weights to reduce impact on fusion results in confidence fusion. Weight values of $w_{c,x}$ and $w_{r,y}$ are given in Fig. 8. In the longitudinal direction, the camera estimate weight $w_{c,x}$ decreases with longitudinal distance increasing. In the lateral direction, radar estimate weight $w_{r,y}$ decreases with lateral distance extending to roadsides. The calculation processes of confidence fusion for camera and radar are presented in Algorithm 1.

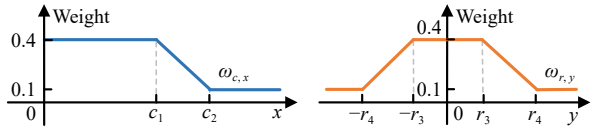


Fig. 8. Confidence weight of the camera and radar in estimate fusion.

Algorithm 1 Calculation Steps of Confidence Fusion for Camera and Radar

Step 1. Initialization

a) Initial state $\hat{\xi}_{l,c}(0)$ and covariance $P_{l,c}(0)$ of local camera estimator is given.

b) Initial state $\hat{\xi}_{j,r}(0)$ and covariance $P_{j,r}(0)$ of local radar estimator is given.

Step 2. Local estimation

a) Local estimates $\hat{\xi}_{l,c}(k_c)$ and covariance $P_{l,c}(k_c)$ of camera are obtained by (11)−(17) for each target.

b) Local estimates $\hat{\xi}_{j,r}(k_r)$ and covariance $P_{j,r}(k_r)$ of radar are calculated by a GM-PHD filter in (18) and (19).

Step 3. Target matching

a) Radar estimates $\hat{\xi}_{j,r}(k_r)$ are aligned to camera sampling times of by (20).

b) Matching results of the camera estimate $\hat{\xi}_{c,n}(k_c)$ and the radar estimate $\hat{\xi}_{r,m}(k_c)$ are obtained by a Kuhn-Munkers algorithm with weight matrix $W^f(k_c)$ in (21).

Step 4. Confidence fusion of detection targets

a) For the targets only detection by the camera, mass values of the camera with "Detected target case" in Table I and the mass values of the radar with " Undetected target case" in Table II are used for fusion.

b) For the targets only detection by the radar, mass values of the camera with "Undetected target case" in Table I and the mass values of the radar with " Detected target case" in Table II are used for fusion.

c) For the target detection by both the camera and the radar, mass values of the camera with "Detected target case" in Table I and the mass values of the radar with "Detected target case" in Table II are used for fusion.

d) Fusion mass value is calculated by the combination rule in (28) and (29).

e) Identification results are obtained by (32)−(34).

Step 5. Confidence fusion estimation

a) Confidence estimation fusion of matched targets are obtained by the weighted maximum likelihood estimation in (35).

Step 6. Set $k_c = k_c + 1$, $k_r = k_r + 1$ and go to Step 2.

Remark 4: For clarity, we reorganize the workflow process of the proposed method as follows:

1) The target estimates in the camera sensor are obtained based on the Kalman filter and the detail is described in Section III-A and Step 2(a) in Algorithm 1. The target estimates in the radar sensor are obtained based on the GM-PHD filter and the detail is described in Section III-B and Step 2(b) in Algorithm 1.

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2) The radar estimates are aligned to camera sampling time with system model extrapolation and the Kuhn-Munkers method is used to obtain the matching relationship of local camera and radar estimates for fusion. This part is presented in Section III-C and Step 3 in Algorithm 1.

3) After matching, three types of targets: targets only detected by the camera, targets only detected by the radar and targets detected by both camera and radar are obtained. Based on the detection results of the camera and radar, confidence fusion is performed one by one to determine whether each target exists. This part is presented in Section III-D and Step 4 in Algorithm 1, and the detailed process is shown in Fig. 7.

4) For the existing targets detected by both camera and radar, states of matched targets are obtained by fusing local estimates of the camera and radar to improve estimation accuracy. The fusing estimation is shown in Section III-E and Step 5 in Algorithm 1.

IV. Experiment Results

In the experiment, an urban road is selected as the experimental area which contains a straight section and a turn section. The camera and radar sensors in Fig. 1 are utilized to collect target information. The initialization and model parameters in local estimators are $Q_{l,c} = Q_r = 0.1I_4$, $R_{l,c} = 0.5I_2$, $R_r = 5I_2$, $d^f = 20$, $\varepsilon_1 = 0.1$, $\varepsilon_2 = 0.3$, $\xi_{l,c}(0) = [x_c(0), 10, y_c(0), 0.06]^T$, $\xi_{j,r}(0) = [x_r(0), 10, y_r(0), 0.06]^T$, $P_{l,c}(0) = P_{j,r}(0) = \text{diag}\{5,5,4,4\}$. Note that $\{x_c(0), y_c(0)\}$ and $\{x_r(0), y_r(0)\}$ are target positions from the camera sensor at time $k_c = 0$ and the radar sensor at time $k_r = 0$, respectively. The confidence area parameters and confidence values are given in Table III.

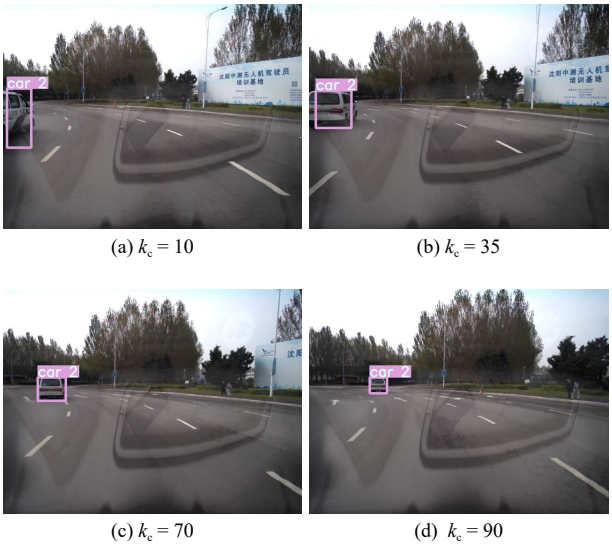
TABLE III
Parameters of Confidence Area for Camera and Radar

Parameter	Value (m)	Parameter	Value
c_1	40	b_{c0}	0.8
c_2	60	b_{c1}	0.8
r_1	−20	b_{c2}	0.2
r_2	−10	b_{r0}	0.8
r_3	10	b_{r1}	0.8
r_4	20	b_{r2}	0.2

A. Performance Comparison in Different Road Scenarios

1) Turning Road Section: The tracking results in a turning road section are given firstly to show the ability to eliminate false targets based on the proposed confidence fusion framework. The camera images in a turning section are shown in Fig. 9. It is shown in Fig. 9 that there exists one target on the road and its ID is Car 2. The original camera data and the local trajectory of Car 2 are given in Fig. 10. The original radar data and local trajectories of the radar are shown in Fig. 11. As shown in Fig. 11, six trajectories are obtained from the radar sensor. However, only radar trajectory 1 represents the trajectory of Car 2, and radar trajectories 2−6 are false target trajectories and other discrete points are clutters in radar measurements.

By the proposed confidence fusion method, two fusion trajectories are obtained which are shown in Fig. 12. Note that the fusion trajectory 1 corresponds to the trajectory of Car 2. The fusion trajectory 2 does not correspond to a real target which is a false target trajectory. The false target in radar reflects the " target existence" with high confidence and reflects " target non-existence" with low confidence in the



(a) $k_c = 10$ (b) $k_c = 35$

(c) $k_c = 70$ (d) $k_c = 90$

Fig. 9. The camera images of the turning road section in some beats.

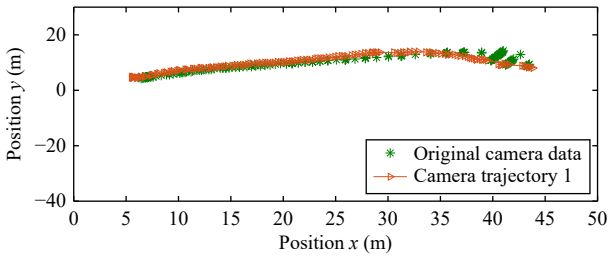


Fig. 10. The original camera data and local trajectory of the camera.

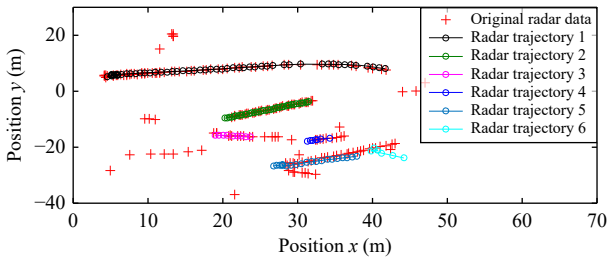


Fig. 11. The original radar data and local trajectories of the radar.

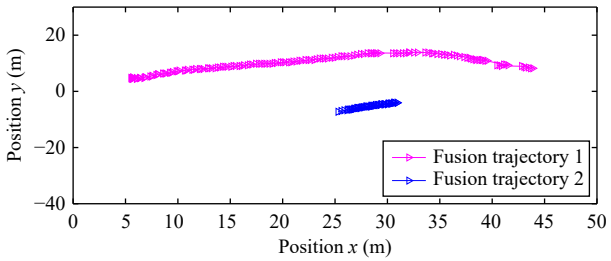


Fig. 12. Fusion trajectories from the proposed confidence fusion method.

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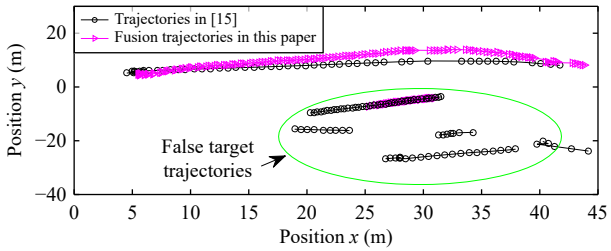


Fig. 13. Fusion trajectories comparison in turning road section.

camera. It is displayed as target existence after fusion, and the false target trajectory 2 is retained in the fusion result. This situation requires the introduction of new sensor information, such as adding new camera sensors or LiDAR sensors, to further determine whether there is a real target at this location.

Moreover, the fusion trajectories comparison is presented in Fig. 13 for our confidence fusion method and the method in [15]. The proposed confidence fusion method with one false target has a smaller false target number compared with [15] with five false targets. In [15], the camera and radar results are directly merged and used as the fusion results, which leads to the false targets in the radar being incorporated into the fusion results. In the proposed method, the confidence fusion is performed sequentially to determine whether each target exists. For the false target in the radar sensor, the mass values of the "Detected target case" in the radar sensor and the mass values of the "Undetected target case" in the camera sensor are used for fusion. Note that the false target is presented in the roadside area, the confidence value of the target existence in the radar is low according to (26). Besides, the confidence value that the false target does not exist in the camera is high based on (25). By fusing the confidence values of the radar and camera, the fusion result shows that the target does not exist. In summary, the proposed method can utilize the low false detection performance of the camera sensor within a short longitudinal distance to eliminate the false targets in the radar during fusion.

2) Straight Road Section: In this part, the confidence fusion results in a straight road section are given to mainly show complementary detection of the camera and radar. The camera images in a straight section are shown in Fig. 14. There exist two targets on the road while only Car 12 is obtained in the camera images. Note that the truck in the right lane is miss detected by the camera. The comparison of the original camera and radar data as well as the fused results are given in Fig. 15. The original radar data include measurements of Car 12, the truck and a roadside obstacle. As shown in Fig. 15, radar results can complement camera detection results but false target measurements are brought at the same time. By the designed confidence fusion strategy, two fused target trajectories are obtained containing Car 12 and the truck. The fusion trajectories of our confidence fusion method and [15] are shown in Fig. 16. The trajectories of Car 12 and the truck are obtained in [15] by complementing the detection targets of the camera and radar. However, the fusion trajectories of [15] contain one false target from the radar by referring to Fig. 14. The fusion results show that the confidence fusion of the camera and the radar complements camera detection results and



Fig. 14. The camera images of the straight road section in some beats.

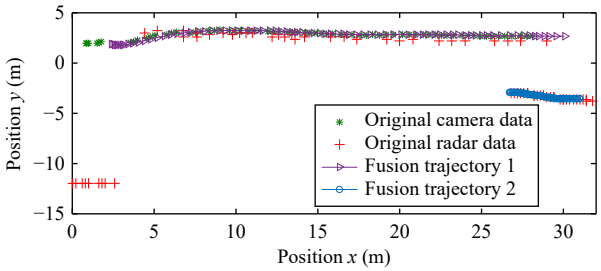


Fig. 15. The position detection results in the straight road section.

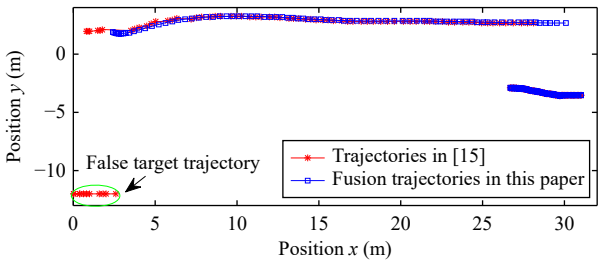


Fig. 16. Fusion trajectories comparison in the straight road section.

eliminates false targets from the radar meanwhile.

To demonstrate the accuracy of confidence estimation of the matched target, a higher precision LiDAR sensor is introduced in the experiment, and the target trajectory from the LiDAR is treated as a true reference value in the comparison. The LiDAR in the experiment is shown in Fig. 17, which is a 125-wire solid-state LiDAR and the model is RS-LiDAR-M1. For clarity, the target positions in longitudinal and lateral directions of matched target Car 12 are given in Fig. 18 and Fig. 19, respectively. The fused results of the proposed method are closer to the truth reference value of the LiDAR sensor compared to the method in [15]. It is shown that the weighted confidence fusion is able to take advantage of high accuracy in the lateral direction of the camera and the longitudinal direction of the radar.

Next the detection results of 400 beats with 715 targets are presented in Table IV to further show the advantages of the

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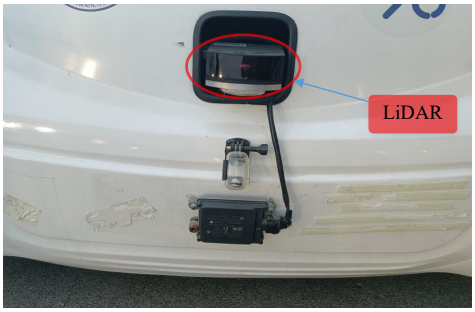


Fig. 17. The LiDAR sensor in the experiment.

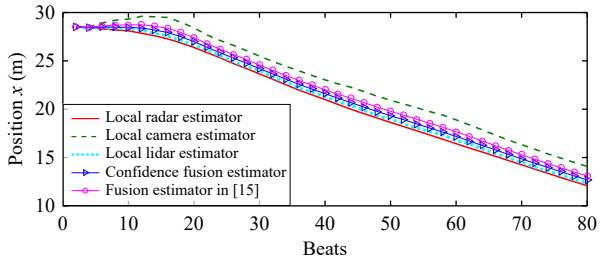


Fig. 18. The longitudinal positions of Car 12.

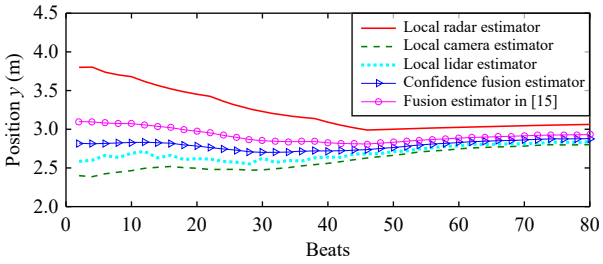


Fig. 19. The lateral positions of Car 12.

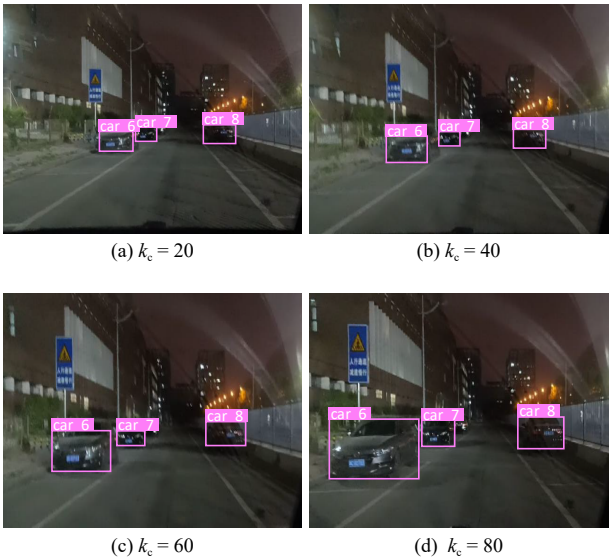
TABLE IV
DETECTION PERFORMANCE COMPARISON OF DIFFERENT METHODS

Method	Miss detection number	Miss detection rate (%)	False target number	False detection rate (%)
Camera sensor	176	24.62	0	0
Radar sensor	83	11.61	157	21.95
The method in [15]	61	8.53	157	21.95
The method in this paper	**61**	**8.53**	**79**	**11.05**

proposed confidence fusion method. The fusion method in [15] decreases the miss detection rate by directly fusing detection results of the camera and radar. The miss detection rate and the false target rate are both reduced by our fusion method by considering the confidence degree of the camera and radar. Specifically, the false detection rate decreased by 50.3% compared to the original radar sensor and the method in [15] during the fusion process.

B. Performance Presentation With Poor Light Conditions

In this subsection, a performance comparison between the proposed confidence fusion method and [14] is presented with poor light conditions. The experiment is conducted at night with only street light illumination. The camera images in poor light condition are shown in Fig. 20 for some beats. The camera and radar data as well as the local trajectories are given in Fig. 21. Note that the camera only detects targets within 35 meters in the longitudinal direction due to poor lighting. The radar is still able to detect targets with long longitudinal distances in the poor light scene. The fused trajectories of the proposed method and [14] with poor light are given in Figs. 22 and 23, respectively. The correspondence between the fusion trajectories in Fig. 22 and the vehicle targets is given in Fig. 24. Based on camera detection results in Fig. 24, it can be seen that the corresponding target in fusion trajectory 4 is missed detection by the camera. The radar detected the corresponding target in fusion trajectory 4 shown in Fig. 21. From Figs. 22 and 23, it is shown that the confidence fusion method and [14] both extend the camera detection range combined with the radar sensor results. However, fusion trajecto-



(a) $k_c = 20$ (b) $k_c = 40$

(c) $k_c = 60$ (d) $k_c = 80$

Fig. 20. The camera images of poor light condition in some beats.

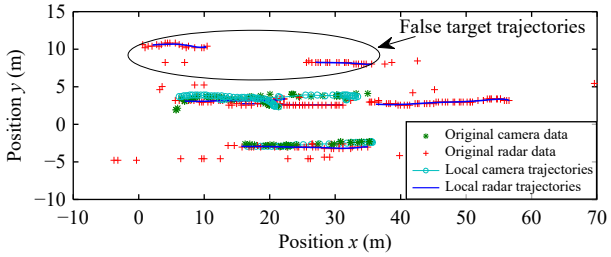


Fig. 21. Trajectories comparison of camera and radar with poor light.

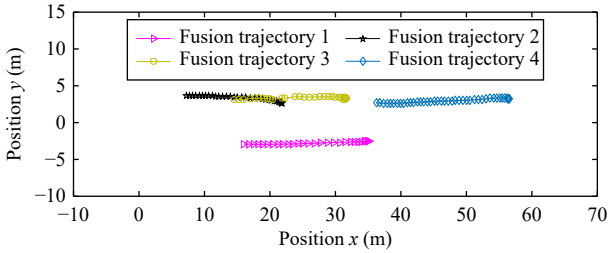


Fig. 22. Fusion trajectories of the proposed method with poor light.

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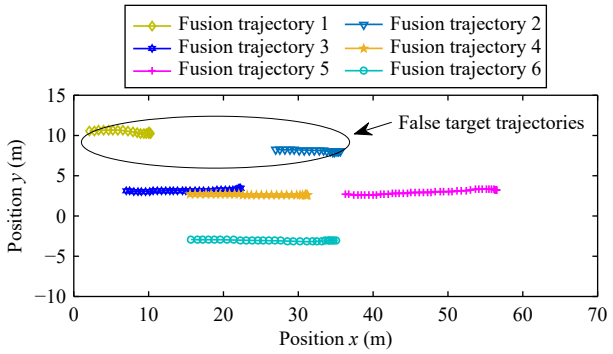


Fig. 23. Fusion trajectories of [14] with poor light.

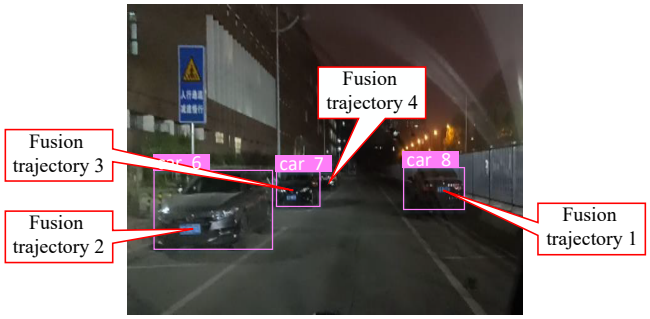


Fig. 24. Correspondence between fusion trajectories and vehicle targets.

ries of [14] contain two false targets from the radar by referring to Fig. 24. Compared with the method in [14], the proposed method utilizes the low false detection performance of the camera within short longitudinal distances, thereby the false targets from the radar sensor are eliminated in the confidence fusion.

Generally speaking, the advantage of the proposed confidence fusion method is to reduce the false target number. This is because the proposed confidence fusion method utilizes the complementary detection performance of the camera with a low false detection rate and the radar with a low miss detection rate. Therefore, by using the long detection range of the radar sensor and the low false detection rate of the camera, the detection range is expanded and the false target number is reduced meanwhile in the proposed confidence fusion method.

C. Robustness to Confidence Parameters

To test the robustness of the proposed fusion method to confidence parameters, a comparison of fusion performance is performed with multiple sets of confidence degree values. Note that a major contribution of this paper is to eliminate false targets in radar by confidence fusion strategy. The fusion result is provided for a false target detected by the radar. The false target is selected in the straight road section and the target position is $(2.6, -12.7)$m shown in Fig. 15. The target is only detected by the radar and not detected by the camera. Hence, mass values of the "Undetected target case" in the camera and mass values of the "Detected target case" of the radar are used in the fusion process. The multiple sets of confidence degree values and fusion results are shown in Table V, in which fusion mass values of target existence event $m(A)$, target nonexistence event $m(\bar{A})$ and uncertain event $m(\{A,\bar{A}\})$ are given. As shown in Table V, the fusion results under different confidence degree values are target nonexistence event $\bar{A}$. That is, we can obtain that the target is a false target under different confidence degree values, which indicates that the proposed confidence fusion strategy is robust to confidence degree parameters.

TABLE V
FUSION RESULTS FOR MULTIPLE SETS OF CONFIDENCE DEGREE VALUES

Parameters	b_{c1}	b_{c2}	b_{r1}	b_{r2}	$m(A)$	$m(\bar{A})$	$m(\{A,\bar{A}\})$	Fusion results
Set of values 1	0.9	0.2	0.9	0.2	0.109	0.802	0.089	$\bar{A}$
Set of values 2	0.8	0.2	0.8	0.2	0.167	0.667	0.167	$\bar{A}$
Set of values 3	0.7	0.2	0.7	0.2	0.197	0.562	0.241	$\bar{A}$
Set of values 4	0.8	0.3	0.8	0.3	0.196	0.643	0.161	$\bar{A}$
Set of values 5	0.8	0.1	0.8	0.1	0.141	0.687	0.172	$\bar{A}$

V. CONCLUSION

A confidence information fusion problem has been addressed for cameras and radars in this paper. Local estimators of the camera and radar have been developed at each sampling time by characteristics of sensor measurements. Local estimation results have been matched by a Kuhn-Munkers algorithm through synchronizing local radar estimates to local camera estimates with a high sampling rate. The mass degrees of the camera and radar have been modeled in the longitudinal and lateral directions, respectively. Then a confidence fusion strategy has been constructed based on the mass degrees of the camera and radar by the D-S evidence theory. A weighted maximum likelihood fusion estimator has been designed for matched targets based on priori positing accuracy of local camera and radar sensors. Finally, the tracking experiments for road vehicles have been performed to present the detection range is expanded and the false target number is reduced meanwhile in the proposed confidence fusion method. Specifically, the false detection rate decreased by 50.3% compared to the original radar sensor and the method in [15] during the fusion process.

For future work, we will focus on the topic of joint target matching and fusion. The camera and radar sensors usually have sensor biases due to their detection principle or environmental interference. The sensor biases seriously affect the accuracy of target matching between two sensors in dense target scenes. The method of joint target matching and fusion with sensor biases is expected to further improve the fusion performance.

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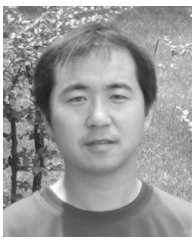
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Xiaohui Hao received the Ph.D. degree in automation from Northwestern Polytechnical University in 2021. From 2021 to 2023, she was a Postdoctoral Research Associate with the School of Automation, Beijing Institute of Technology. She is currently a Lecturer with the School of Artificial Intelligence, Tiangong University. Her current research interests include estimation theory, target tracking, and multi-sensor fusion.



Yuanqing Xia (Fellow, IEEE) received the M.S. degree in fundamental mathematics from Anhui University in 1998 and the Ph.D. degree in control theory and control engineering from Beijing University of Aeronautics and Astronautics in 2001. During January 2002 to November 2003, he was a Postdoctoral Research Associate with the Institute of Systems Science, Academy of Mathematics and System Sciences, Chinese Academy of Sciences. From November 2003 to February 2004, he was with the National University of Singapore as a Research Fellow, where he worked on variable structure control. From February 2004 to February 2006, he was with the University of Glamorgan, Pontypridd, UK, as a Research Fellow. From February 2007 to June 2008, he was a Guest Professor with Innsbruck Medical University, Innsbruck, Austria. Since 2004, he has been with the School of Automation, Beijing Institute of Technology first as an Associate Professor, then, since 2008, as a Professor. His current research interests include networked control systems, robust control and signal processing, and active disturbance rejection control.



Hongjiu Yang received the Ph.D. degree in control science and engineering from Beijing Institute of Technology in 2011. He was an Associate Professor with the Department of Automation, Institute of Electrical Engineering, Yanshan University from 2011 to 2018, where he has been a Lecturer since 2011 and an Associate Professor since 2013. He is currently a Professor with the School of Electrical and Information Engineering, Tianjin University. His current research interests include robust control/filter theory, networked control systems, model predictive control, and active disturbance rejection control.



Zhiqiang Zuo (Senior Member, IEEE) received the M.S. degree in control theory and control engineering in 2001 from Yanshan University and the Ph.D. degree in control theory in 2004 from Peking University. In 2004, he joined the School of Electrical and Information Engineering, Tianjin University, where he is a Full Professor. From 2008 to 2010, he was a Research Fellow in the Department of Mathematics, City University of Hong Kong, China. From 2013 to 2014, he was a Visiting Scholar at the University of California, Riverside, USA. His research interests include nonlinear control, robust control and multi-agent systems. Dr. Zuo is an Associate Editor of the *Journal of the Franklin Institute*.

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